ECE 340: Semiconductor Devices

Lecture 7: Fermi distribution & Fermi level

Prof. Hyunseok Kim

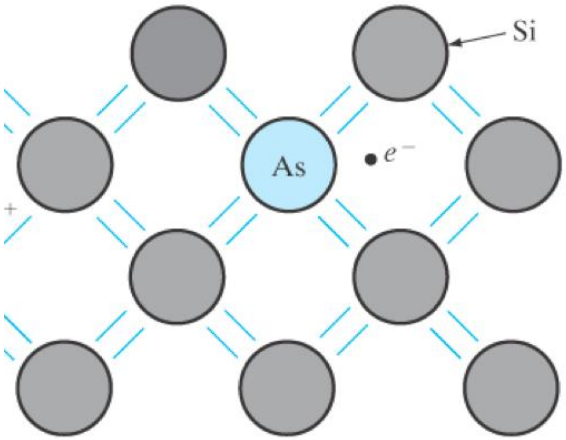
Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

E-mail: hyunseok@illinois.edu

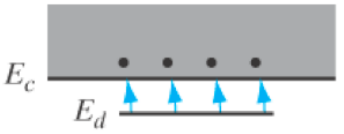
Office Hours: Thursdays 12p-1p, 2256 HMNTL

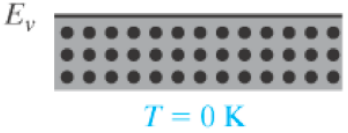
(Recap) Extrinsic Material

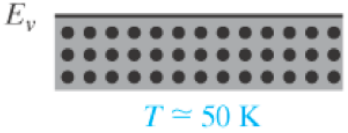
As impurity (donor)



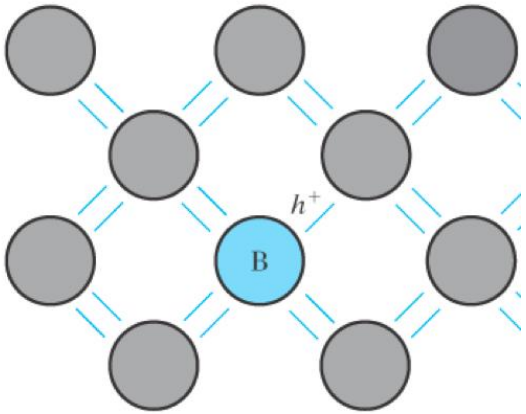






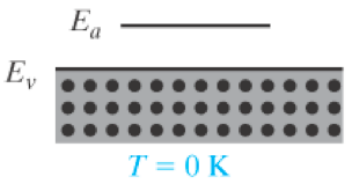


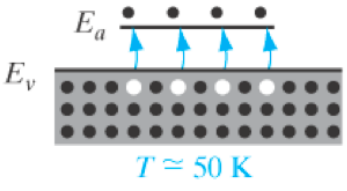
B impurity (acceptor)











From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

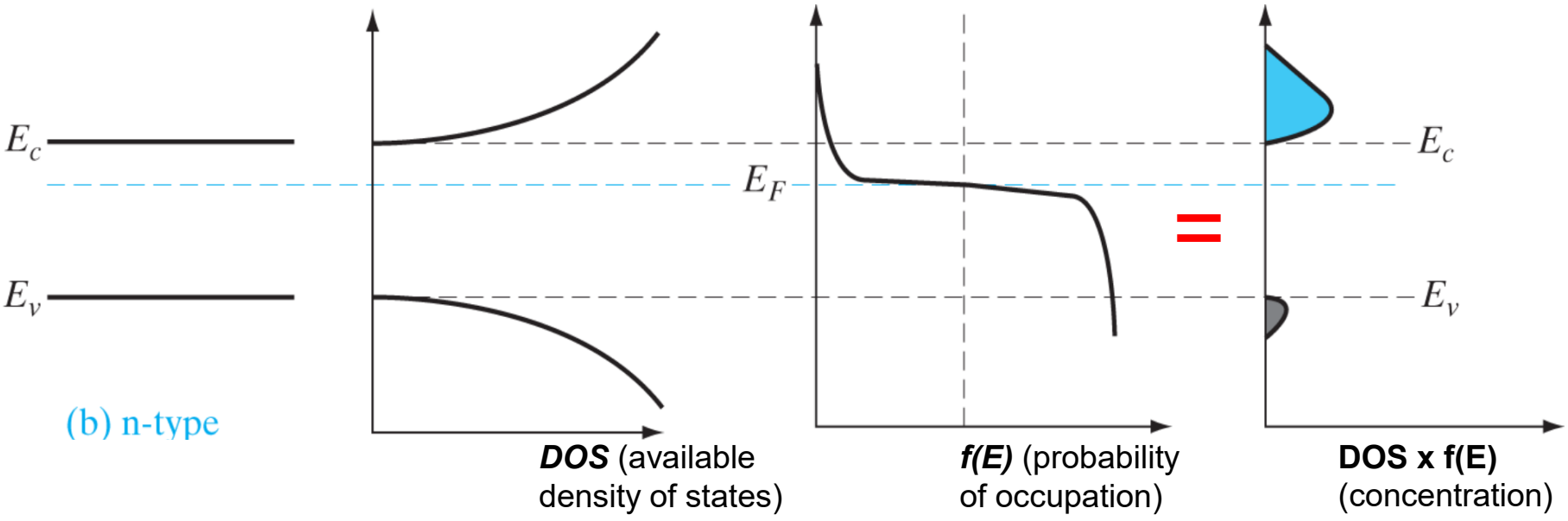
(Recap) Important Definitions

- **Intrinsic Material:** For a "perfect" crystal in equilibrium, electron and hole densities are equal
  - The value of this density n_i is called the intrinsic concentration
- **Extrinsic Material:** The crystal is "doped" with
  - **donors**, where the material is n-type and the electron density exceeds the hole density, or
  - **acceptors**, where the material is p-type and the hole density exceeds the electron density
  - An **amphoteric dopant** can act as either a donor or acceptor, depending upon what atom it is replacing in the crystal
  - In extrinsic material, the carrier with the higher concentration is the **majority carrier** and the carrier with lower concentration is the **minority carrier**
    - We will find out later that the product of the minority carrier density and majority carrier density is constant for a given temperature (in equilibrium), and equal to n_i^2
    - In p-type material, holes are the majority carrier, electrons the minority carrier
    - In n-type material, electrons are the majority carrier, holes the minority carrier

(Recap) calculating carrier density

- Integration of [The number (density) of states at a given energy] x [the probability a state is filled at that energy]

$$n_0 = \int_{E_c}^{\square} f(E) N_c(E) dE$$



Fermi-Dirac Distribution

- The probability that an available energy state at E will be occupied by an electron at temperature T (temperature in Kelvin) is given by the **Fermi-Dirac Distribution Function**
- Fermi-Dirac distribution function:

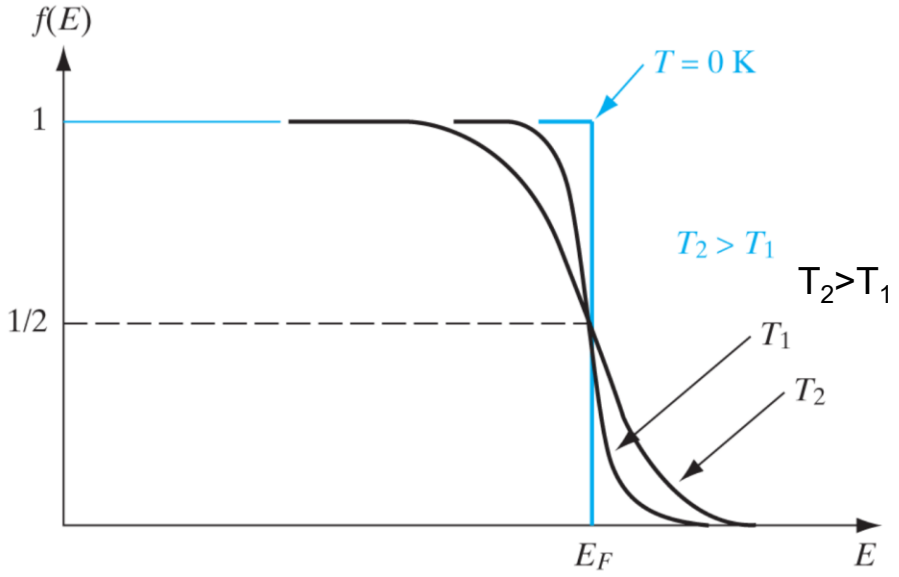
$$f(E) = \frac{1}{1 + e^{(E - E_f)/kT}}$$

E_f : Fermi Level

k : Boltzmann Constant

$$k = 8.62 x 10^{-5}\, eV / K$$

- **Fermi level (E_f)**: the energy that determines how likely an electronic states is to be occupied by an electron.
- The Fermi level has units of energy (typically eV)



(a) $E = E_f$ $\quad f(E = E_f) = \frac{1}{1 + e^0} = \frac{1}{2}$

(b) T = 0K

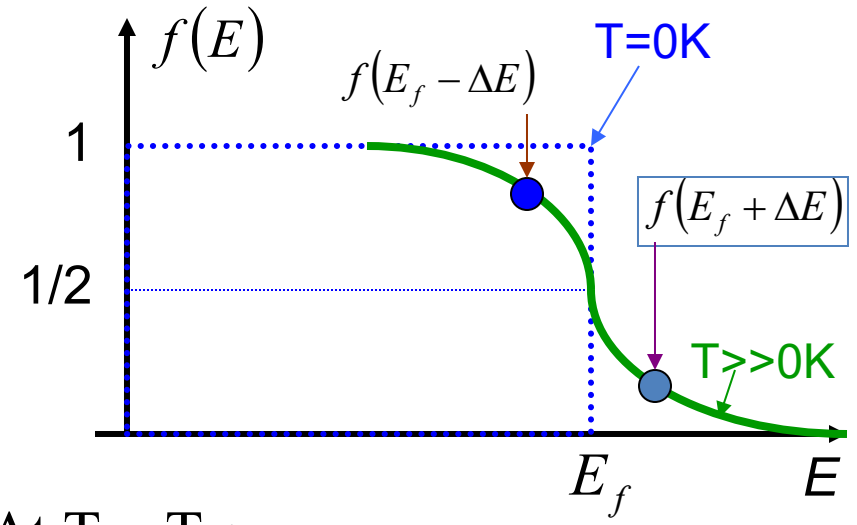
$$f(E < E_f) = \frac{1}{1 + e^{-\square}} = 1$$

$$f(E > E_f) = \frac{1}{1 + e^{+\square}} = 0$$

What About Holes?

Hole Probability:

- A state is either empty or full, so the probability a state is empty is 1-P(full)
- P(full) is given by the Fermi distribution *f(E)*
- P(empty) = 1 - *f(E)*
- $f(E_f+\Delta E)$ is the probability of finding an electron at E+ΔE
- 1-$f(E_f-\Delta E)$ is the probability of finding a hole at E-ΔE

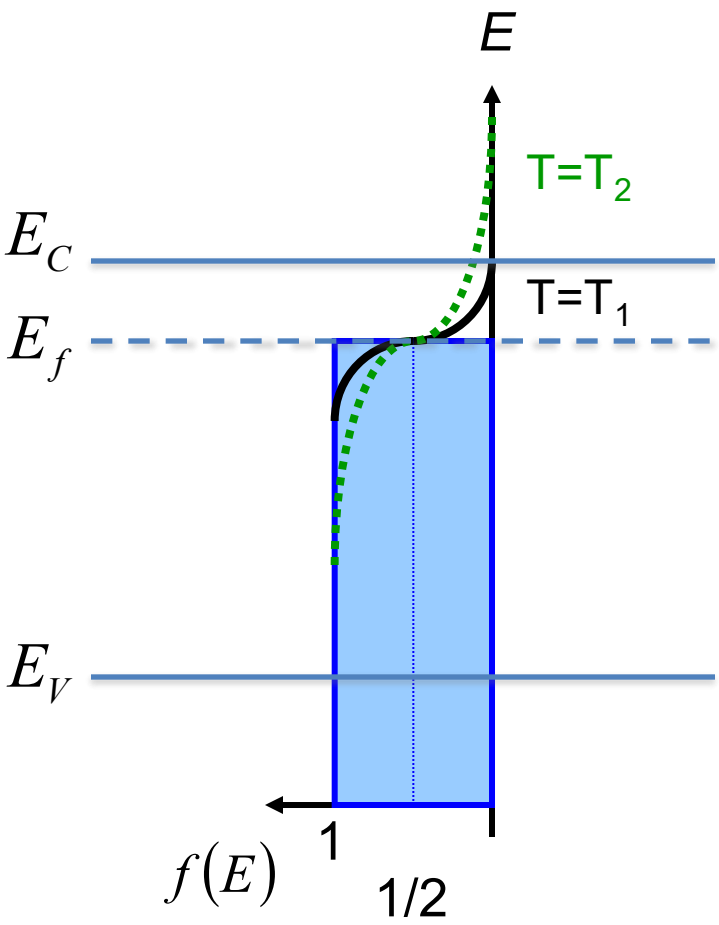


At $T = T_1$:

$$f(E) = \frac{1}{1+e^{(E-E_f)/kT_1}}$$

$$1-f(E) = \frac{e^{(E-E_f)/kT_1}}{1+e^{(E-E_f)/kT_1}} = \frac{1}{1+e^{(E_f-E)/kT_1}}$$

Approximating the Fermi Function



$$f(E) = \frac{1}{1 + e^{(E - E_f)/kT}}$$

$$kT = \left(8.62x10^{-5}\frac{eV}{K}\right)(300K) \sim 0.0259eV$$

In the case where $E = E_C$

$$f(E_C) = \frac{1}{1 + e^{(E_C - E_f)/kT}}$$

Note: $e^1 \sim 2.7,\ e^2 \sim 7.4,\ e^3 \sim 20$

$$\frac{1}{1+e^3} = \frac{1}{21} = 0.048 \sim 0.05 = \frac{1}{20} = \frac{1}{e^3} = e^{-3}$$

So, for $\dfrac{(E_C - E_f)}{kT} > 3$, or $\left(E_C - E_f\right) >\sim 3kT$

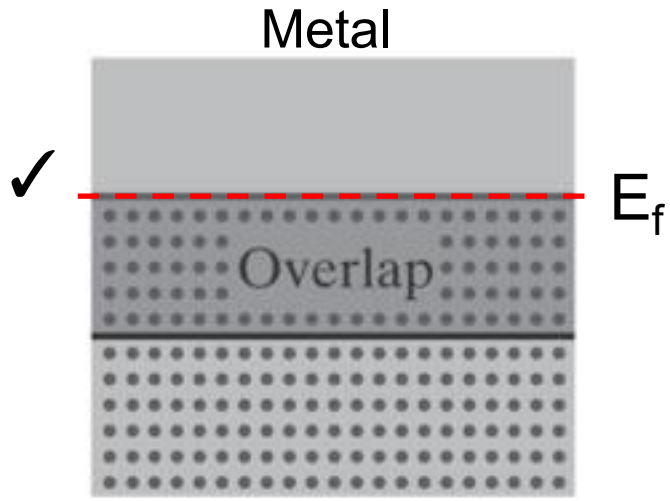
$$f(E_C) = \frac{1}{1 + e^{(E_C - E_f)/kT}} \sim e^{-(E_C - E_f)/kT}$$

(non-degenerate case)

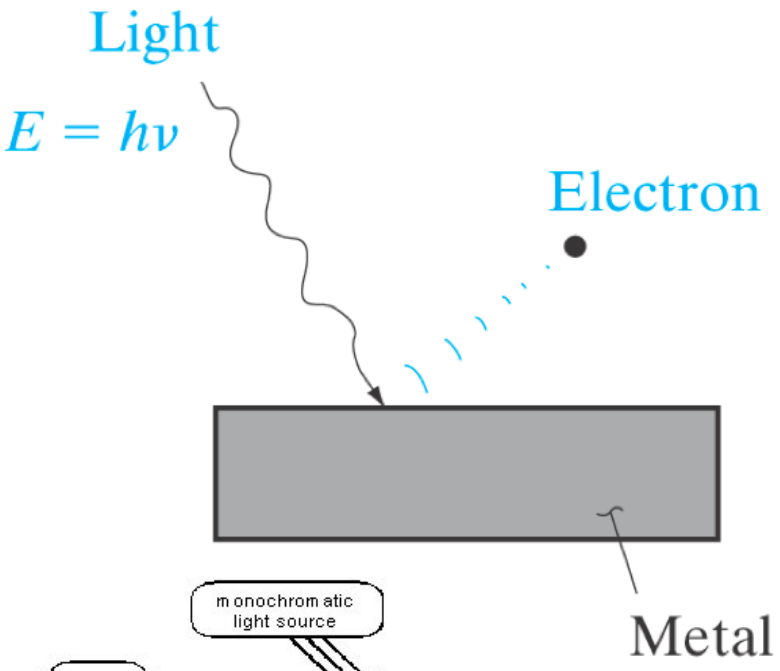
Fermi Level in Metals

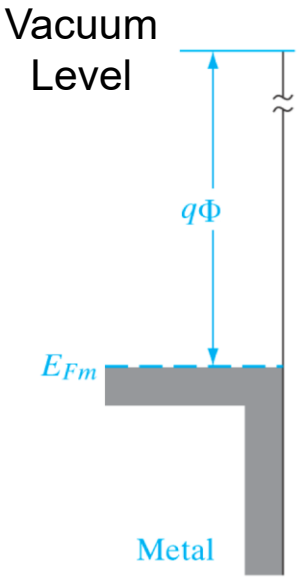
- For metals, the Fermi level is the energy of the topmost filled level in the ground state of an N electron system (implies the temperature is absolute zero, 0K)
  - Related to the energy needed to remove an electron from a metal at absolute zero:

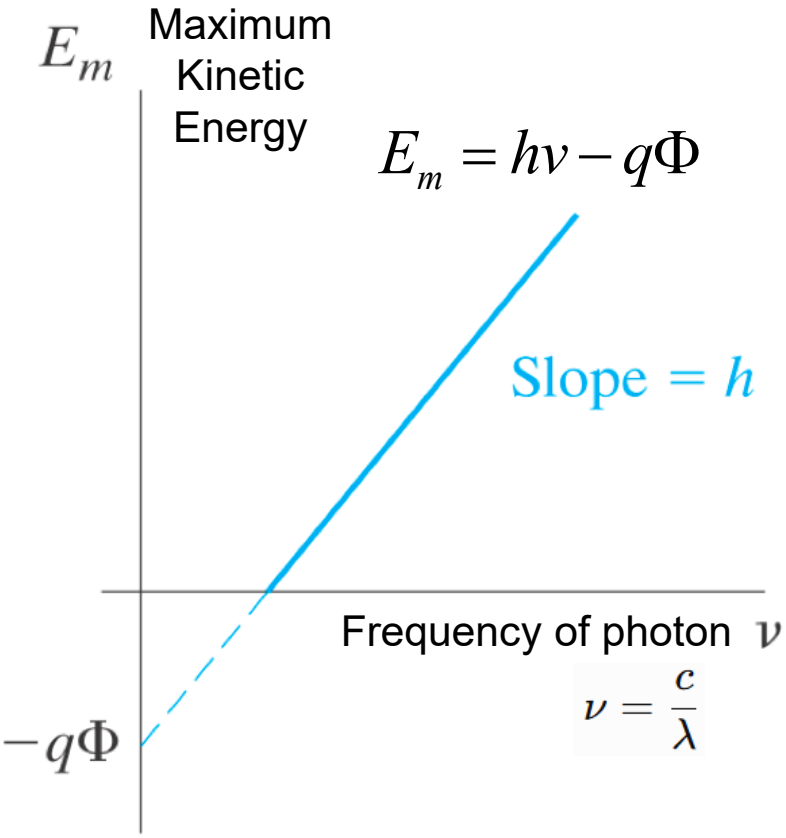
$$\Phi = E_{\text{vac}} - E_F$$

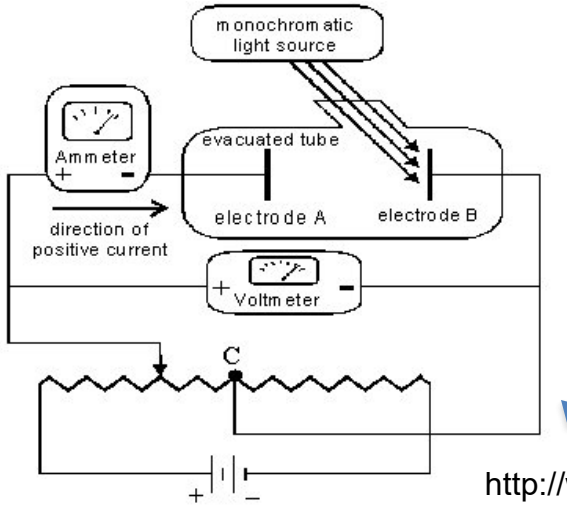


Metal Fermi Energy









- In a metal, the Fermi Energy referenced to the vacuum level is the metal work function

http://www.sci.ccny.cuny.edu/~rstein/papers/jcmst00/jcmst00.html

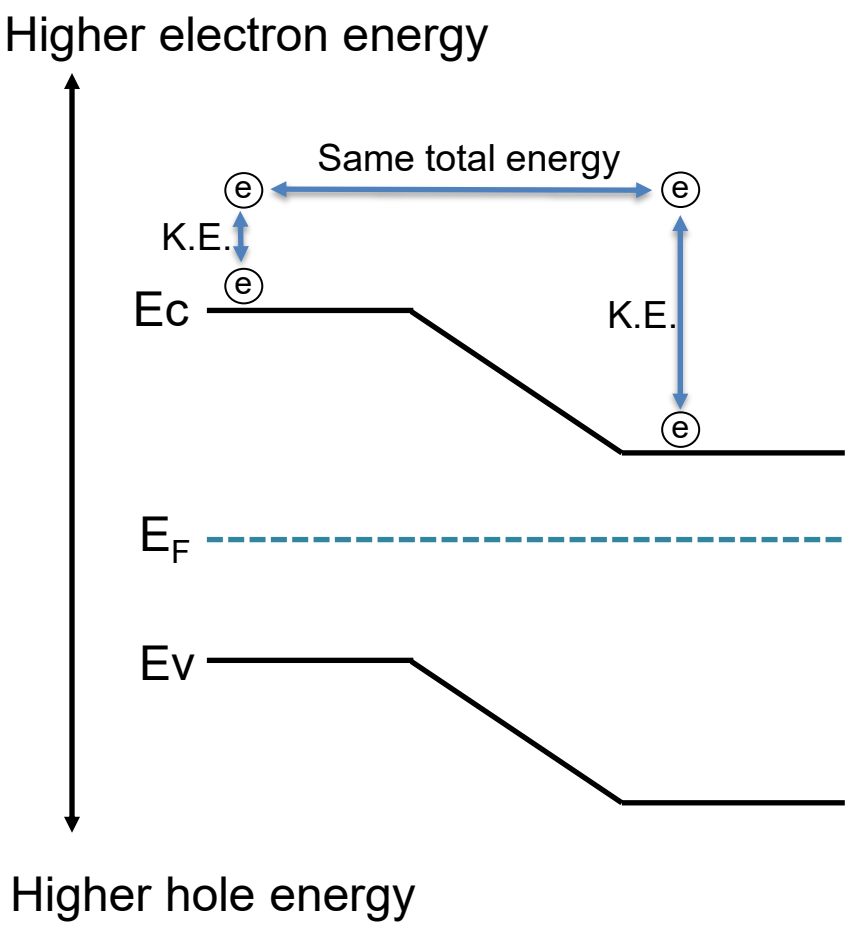
Fermi Level in Semiconductor

- There are several equivalent definitions which can be used:
  - A measure of how difficult for electron to be added or removed
  - The sum of the chemical potential and internal electrostatic potential energy, generally equal to the electrochemical potential
- ***Electrochemical Potential:*** the total energy "preference" for a charged particle to be at a particular location, incorporating both ***chemical potential*** and ***electrostatics (influence of electric field)***
- The ***Fermi Level (or Fermi Energy)*** is equal to the electrochemical potential in a semiconductor.

Fermi Level in Semiconductor

- Comment 1: If material with different Fermi levels are brought together, the Fermi levels must align. This has certain implications which will be discussed soon.
- Comment 2: Detailed understanding of the Fermi level and Fermi energy requires a background in statistical mechanics, which is beyond the scope of this course

Revisiting band diagram



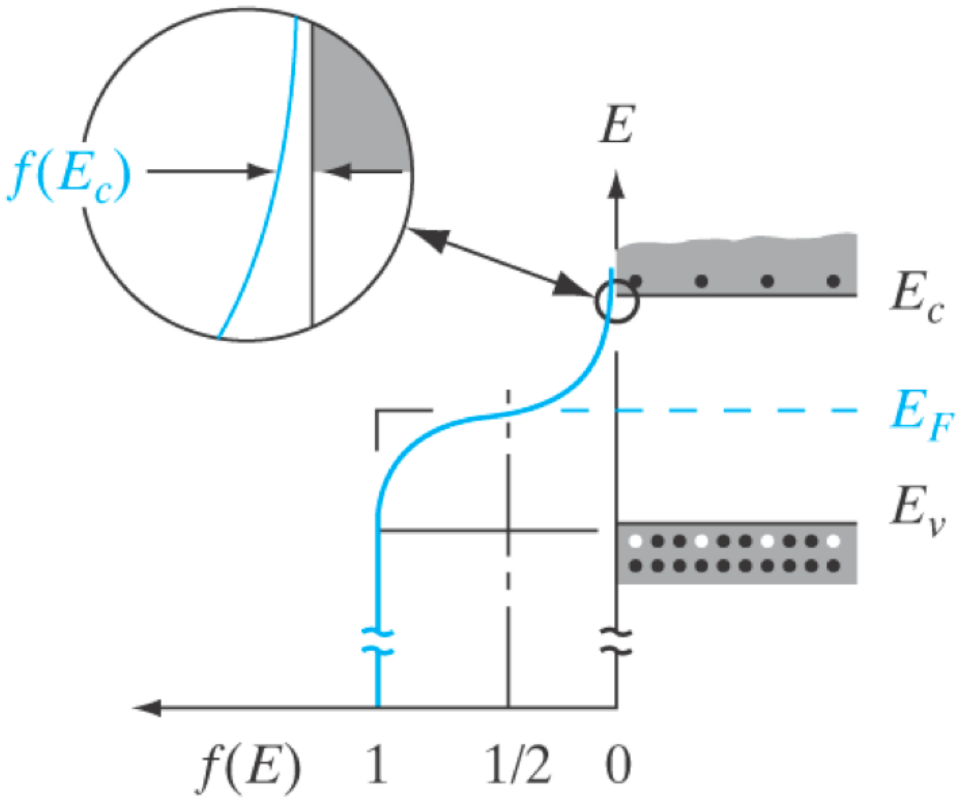
- **Fermi level, E_F**: Electrochemical potential: Chemical potential + electrostatic potential. In isolated system under equilibrium, E_F must align, otherwise there will be net driving force for electrons & holes.
- **E_c & E_v**: Minimum allowed electron (and hole) energy state in the conduction (and valence) band.
- Electron above *Ec* has kinetic energy.
- Hole below *Ev* has kinetic energy.

Fermi Level: Experimental Determination

- In practice, it is relatively straightforward to measure carrier concentration
- The Fermi level is typically calculated from the measured carrier concentration using the relationships discussed last time (non-degenerate case), or slightly more complex relationships for degenerate material

Fermi Level Versus Doping Type

Intrinsic



Remember:

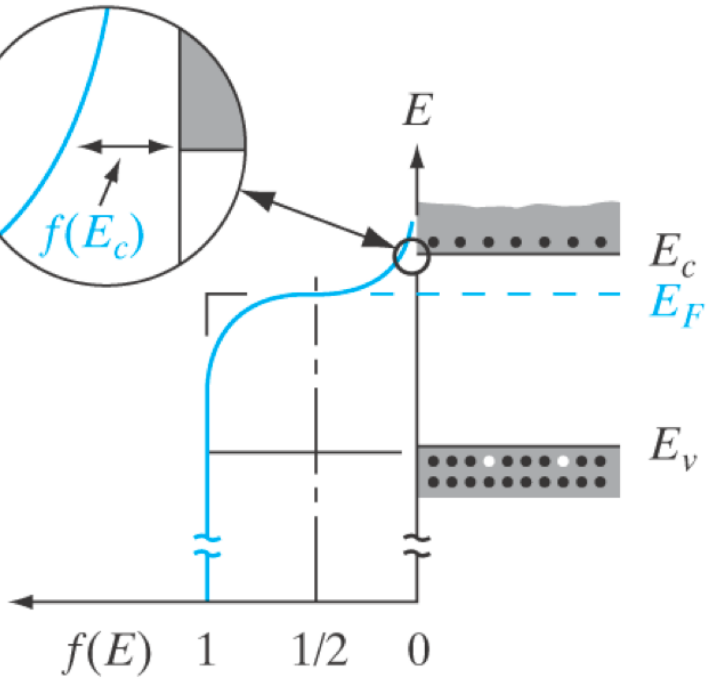
At $T = T_1$:

$$f(E) = \frac{1}{1 + e^{(E - E_f)/kT_1}}$$

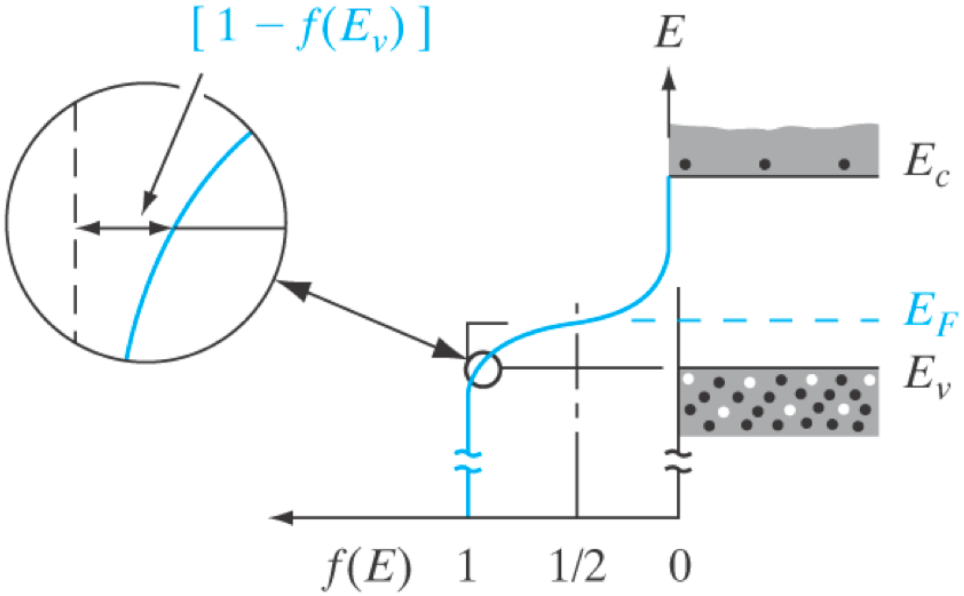
$$1 - f(E) = \frac{e^{(E - E_f)/kT_1}}{1 + e^{(E - E_f)/kT_1}} = \frac{1}{1 + e^{(E_f - E)/kT_1}}$$

Fermi Level Versus Doping Type

N-type

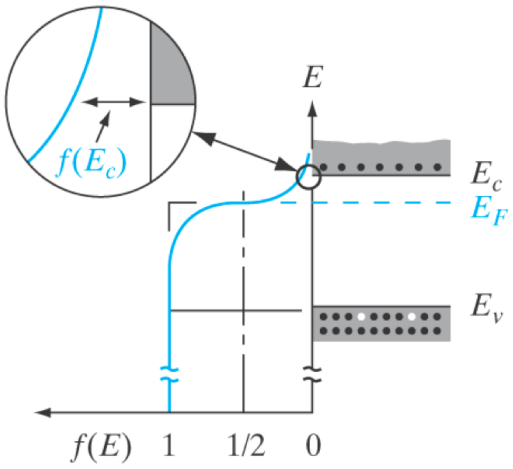


p-Type

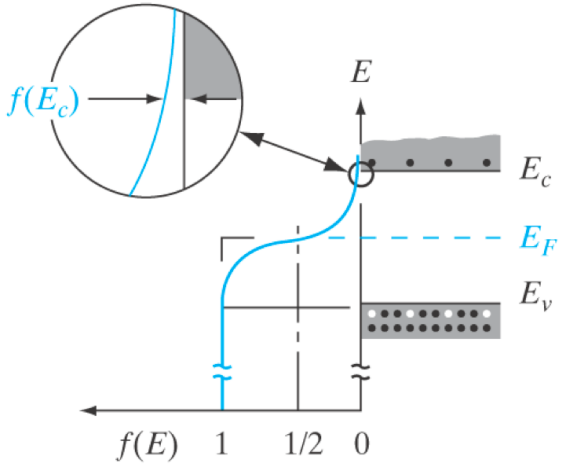


Fermi Level Versus Doping Type

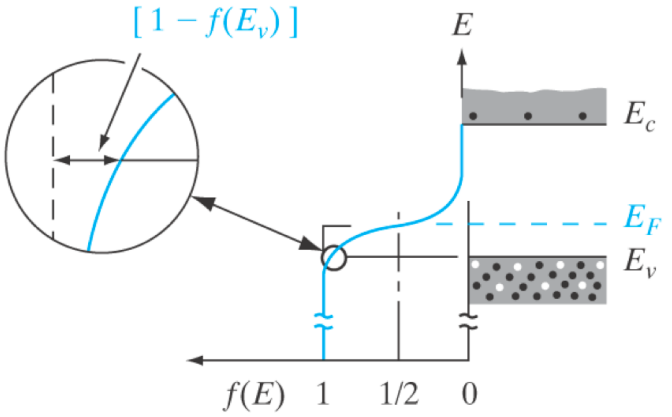
n-Type



Intrinsic



p-Type



Remember: At $T = T_1$:

$$f(E) = \frac{1}{1 + e^{(E - E_f)/kT_1}}$$

$$1 - f(E) = \frac{e^{(E - E_f)/kT_1}}{1 + e^{(E - E_f)/kT_1}} = \frac{1}{1 + e^{(E_f - E)/kT_1}}$$

Density of States Calculation

Additional Information

Conduction Band Density of States

Consider a cube of material having sides of length "L"

A free electron in this material has a wavefunction given by:

$\psi = Ae^{j\left(k_x x + k_y y + k_z z\right)}$ where

$$k_x = n_x \frac{2\pi}{L},\ k_y = n_y \frac{2\pi}{L},\ k_z = n_z \frac{2\pi}{L}$$

We also know:

$$E = \frac{p^2}{2m^*} = \frac{\hbar^2 k^2}{2m^*} = \frac{2\pi^2\hbar^2}{m^* L^2}\left(n_x^2 + n_y^2 + n_z^2\right) = \frac{2\pi^2\hbar^2}{m^* L^2} n^2$$

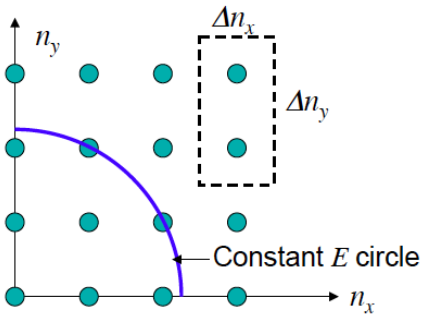
We can also express n as a function of E: $n = \frac{L}{\pi}\sqrt{\frac{m^* E}{2\hbar^2}}$

The constant energy surface is therefore the the surface of a sphere of radius "n" where:

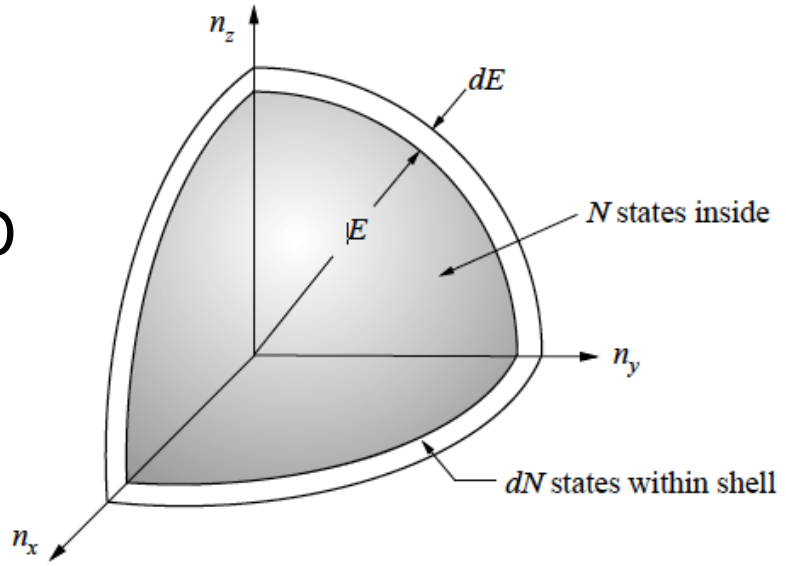
$$n = \left(n_x^2 + n_y^2 + n_z^2\right)^{\frac{1}{2}}$$

The total number of states within the sphere is:

$$N' = \frac{4}{3}\pi n^3$$



2D



3D

Note: Other variants of this derivation get the same result

Conduction Band Density of States (2)

Including spin, the total number of states up to a given energy is therefore:

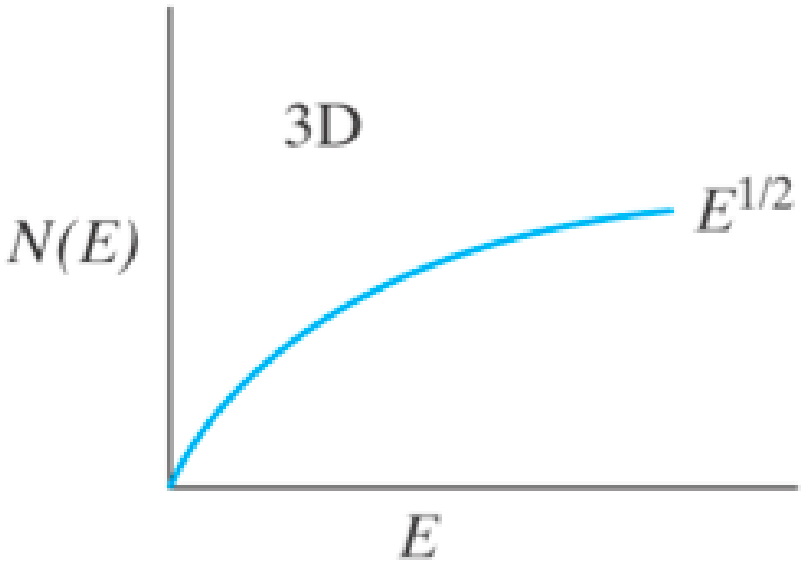
$$N'(n) = 2 \times \frac{4}{3}\pi n^3 = \frac{8}{3}\pi n^3$$

Using the expression for n:

$$n = \frac{L}{\pi}\sqrt{\frac{m^* E}{2\hbar^2}} \quad \text{so:}$$

$$N'(E) = \frac{8}{3\pi^2} L^3 \left(\frac{m^* E}{2\hbar^2}\right)^{3/2}$$

$$\frac{dN'(E)}{dE} = \frac{L^3}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E}$$



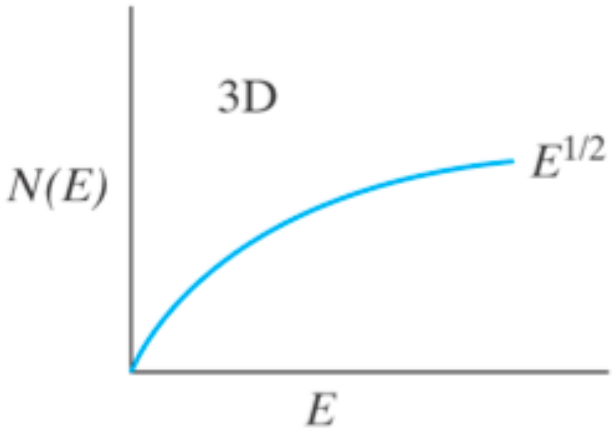
The Density of States is defined as the number of states per incremental unit of energy near E per unit volume:

$$N(E) = \frac{dN'(E)}{dE}\frac{1}{L^3} = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E}$$

Conduction Band Density of States (2)

DoS calculated

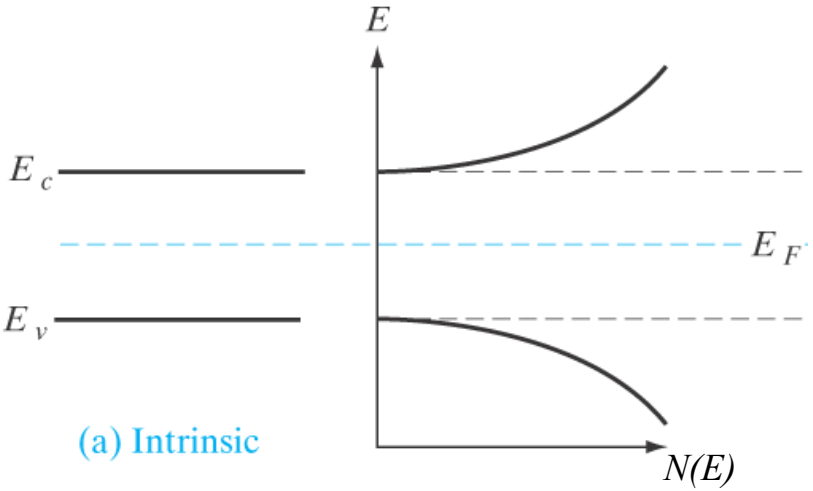
$$N(E) = \frac{dN'(E)}{dE}\frac{1}{L^3} = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E}$$



DoS of electron in conduction band

$$N_c(E) \propto \sqrt{E - Ec}$$

(E-E$_c$) instead of *E* because the energy relative to the band edge (*E$_c$*) is its kinetic energy within the band

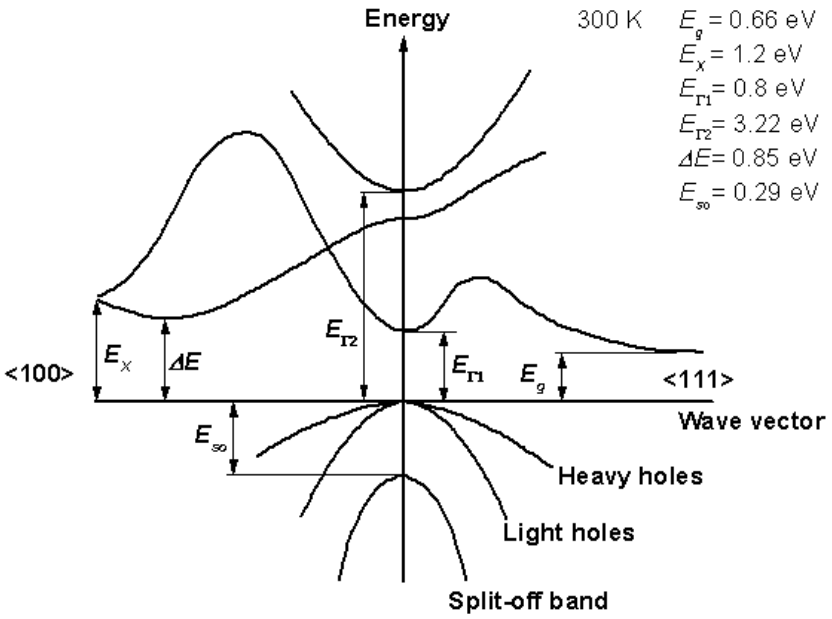


DoS of hole in valence band

$$N_v(E) \propto \sqrt{E_v - E}$$

Other Cases

- Valence Band
  - Need to include both **light holes** and **heavy holes** in the density of states
- Near the Direct-Indirect Crossover
  - Need to include both valleys
- Indirect Materials
  - Multiple valleys due to symmetry and energy degeneracy
  - This is included in the **density of states effective mass**

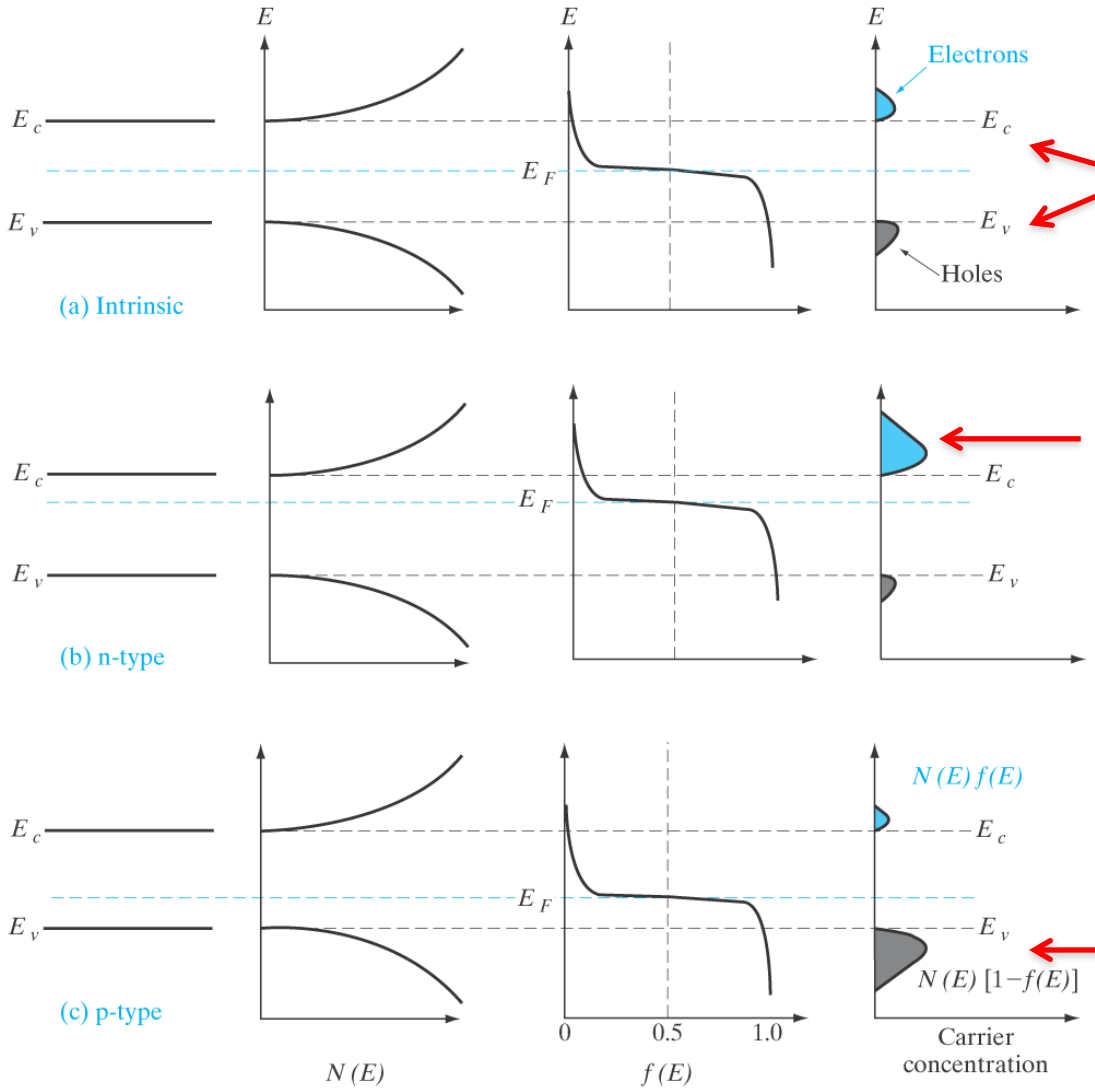


http://www.ioffe.rssi.ru/SVA/NSM/Semicond/Ge/Figs/221.gif

Equilibrium Electron and Hole Concentrations

Carrier Concentration at Thermal Equilibrium

The Fermi Function Multiplied by the Density of States



n_i, p_i: intrinsic concentration

Conduction band DOS

$$n_i = \int_{E_c}^{\infty} f(E) N_c(E) dE$$

$$= p_i = \int_{-\infty}^{E_V} [1 - f(E)] N_V(E) dE$$

Probability the state is empty

Valence band DOS

$$n_0 = \int_{E_c}^{\square} f(E) N_c(E) dE$$

n_0, p_0: equilibrium concentration (could be intrinsic or extrinsic)

Thus, $n_0 = p_0 = n_i$ for intrinsic Si

$$p_0 = \int_{-\infty}^{E_v} [1 - f(E)] N_v(E) dE$$